Surrogate Aerodynamic Modeling: Predicting Transonic Airfoil Coefficients via Machine Learning

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https://github.com/ashishdasu/transonic-aerodynamic-surrogate-modeling

Abstract

High-fidelity CFD simulations of transonic airfoil flow are accurate but require minutes per evaluation on modern hardware, making them impractical for design-space exploration. We train four regression surrogates—polynomial regression, random forest, XGBoost, and a PyTorch MLP—to predict lift coefficient C_l and pitching moment coefficient C_m of eight distinct airfoils at Mach 0.65–0.85, directly from 18 geometric and flight-condition features. All models are trained on the TransonicSurrogate dataset [1] (1,362 OpenFOAM `rhoCentralFoam` runs). We hold out all RAE2822 samples as an unseen-geometry extrapolation probe—a test the source paper never performed. On the interpolation test split, XGBoost achieves $R^2 > 0.997$, matching the paper's best result; on the RAE2822 holdout every model yields negative R^2, revealing that random-split accuracy does not transfer to unseen airfoil families. All surrogates reduce inference time from $\sim$7 min (CFD) to $\lesssim$4 ms—a speedup exceeding $10^5\times$.

1 Introduction

Aircraft design and flight-control synthesis require evaluating aerodynamic force and moment coefficients at thousands of operating points. In the transonic regime ($M_\infty \in [0.6, 0.9]$), shock waves form whose location and strength are strongly nonlinear functions of geometry and flight condition. Accurate transonic analysis requires solving the Euler or Navier–Stokes equations, incurring wall-clock times of minutes per condition even on modern workstations.

Surrogate modeling—replacing expensive simulations with cheap function approximators trained on a pre-computed database—is a practical alternative for design sweeps. The transonic challenge is that smooth regressors struggle near shock-induced nonlinearities while the database must cover both the geometry and flight-condition spaces.

Neural-network surrogates for airfoil aerodynamics have been explored in related work [2], but prior studies target subsonic regimes or treat geometry as a categorical variable rather than a continuous parameterization. Elrefaie et al. [1] introduced the TransonicSurrogate dataset and evaluated GP regression [3], random forest [4], and gradient-boosted trees [5], reporting $R^2 \approx 0.996$ with random train/test splits across all eight airfoils. That result, while impressive, is produced by an evaluation design that mixes the RAE2822 supercritical profile into both training and test data, effectively measuring interpolation within seen geometries rather than generalization to new ones.

This work revisits the same dataset with a more rigorous evaluation and substantially deeper analysis. We withhold all 172 RAE2822 samples from training entirely—a *geometry holdout* that tests generalization to an unseen airfoil family rather than interpolation within known ones. Every model fails

this test ($R^2 < 0$), a result the original evaluation could not reveal. Beyond the holdout, we train four surrogates: a polynomial baseline to quantify where smooth global fits break down, random forest and XGBoost as the tree-based methods the paper used, and a PyTorch MLP [6] with a shared trunk and dual C_l/C_m heads as a fully continuous deep-learning alternative. We omit GP regression (the paper's strongest model) because its $O(n^3)$ scaling conflicts with the millisecond-inference goal of a deployed surrogate. We also run four analyses the paper never performed: leave-one-airfoil-out generalization, data efficiency curves, Mach-regime extrapolation, and feature ablation.

2 Dataset and Problem Setup

2.1 Data Source

The TransonicSurrogate dataset [1] contains 1,362 steady compressible Euler solves computed with OpenFOAM's `rhoCentralFoam` solver on structured O-meshes ($100c$ far-field). Each row is one (airfoil, α, M_∞) triple; the sweep covers:

- **8 airfoils:** RAE2822, RAE5212, NACA0012, NACA2412, NACA4412, NACA23012, NACA24112, NACA25112 (≈ 171 conditions each).
- **5 Mach numbers:** $M_\infty \in \{0.65, 0.70, 0.75, 0.80, 0.85\}$.
- **34 AoA values:** $\alpha \in [-2.0°, +14.5°]$, $0.5°$ steps.

The CSV provides no airfoil-name column; airfoil identity is recovered by a six-coordinate fingerprint $(y_{U_1}, y_{U_4}, y_{U_8}, y_{L_1}, y_{L_4}, y_{L_8})$ matched against a hand-validated lookup table. Figure 1 shows the recovered profiles.

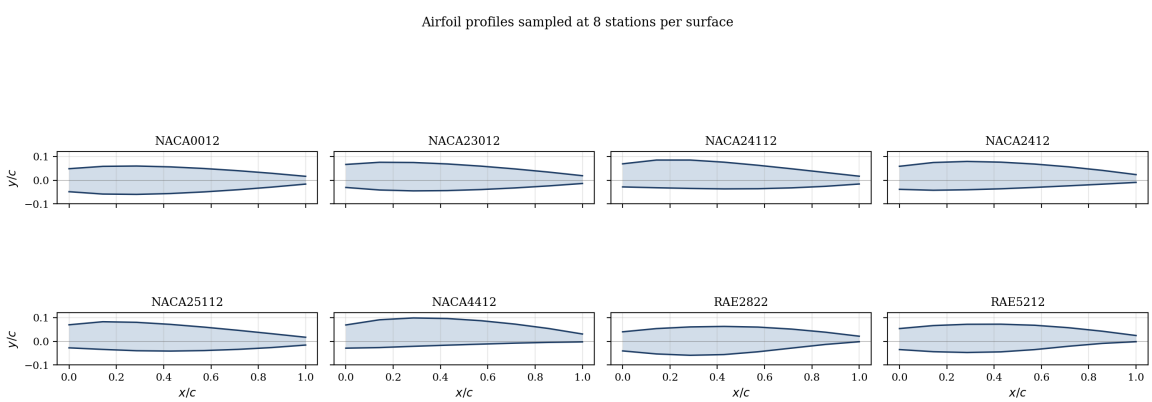

Figure 1: The eight airfoil profiles recovered from the geometry fingerprint. NACA profiles share a family resemblance (varying camber and thickness). RAE2822 and RAE5212 are supercritical designs with a flatter upper surface and reflexed trailing edge—the visual basis for their out-of-distribution behavior when held out.

2.2 Feature Vector and Targets

The 18-dimensional feature vector $\mathbf{x} = [y_{U_1}, \ldots, y_{U_8}, y_{L_1}, \ldots, y_{L_8}, \alpha, M_\infty]$ encodes airfoil shape as a continuous 16-dimensional surface-coordinate parameterization. Airfoil identity is never one-hot encoded; the model must infer geometry solely from the continuous coordinates.

The prediction targets are the lift coefficient C_l (net upward force normalized by dynamic pressure and wing area) and the pitching moment coefficient C_m (the nose-up/nose-down torque about the quarter-chord point). Both are dimensionless and capture the two quantities designers care most about:

will the wing generate enough lift, and is the aircraft stable? This is a two-output regression problem. All inputs and targets are z-score normalized; scalers are fit on the training split only.

2.3 Data Splits

All 172 RAE2822 rows are carved out before any splitting to form a held-out geometry probe. The remaining 1,190 rows (seven airfoils) are split 60/20/20 (train/val/test, seed 42), yielding 714/238/238 samples.

This design deliberately diverges from Elrefaie et al. [1], who pooled all 1,362 samples into a random 80/20 split (RAE2822 in both train and test). Their approach evaluates *interpolation*—generalizing to held-out rows from a *known* geometry. Our geometry holdout evaluates *extrapolation*: can a surrogate predict for a supercritical profile it has *never seen*?

The validation split is used only for hyperparameter selection; both test splits are evaluated exactly once, after all tuning is done.

3 Methods

3.1 Polynomial Regression Baseline

A degree-d polynomial feature expansion (`sklearn.PolynomialFeatures`) is fit with Ridge regression, regularization λ selected by 5-fold CV over $d \in \{2, 3\}$, $\lambda \in \{0.01, 0.1, 1.0, 10.0\}$. The best configuration (degree 3, $\lambda = 1.0$) produces 1,140 basis functions from 714 training samples, making the problem near rank-deficient without regularization. Its inability to handle shock-induced nonlinearity is exactly what motivates the non-parametric models that follow.

3.2 Random Forest

`sklearn` [7] `MultiOutputRegressor` wrapping `RandomForestRegressor` [4] with grid $n_{\text{est}} \in \{50, 100, 200\}$, $d_{\max} \in \{5, 10, \texttt{None}\}$, 5-fold CV (neg-RMSE objective). Best configuration: 200 trees, unlimited depth.

3.3 XGBoost

XGBoost [5] does not natively support multi-output regression; we fit one `XGBRegressor` per target and wrap both in an `XGBoostMultiOutput` class. Grid: $\eta \in \{0.01, 0.05, 0.1\}$, $d_{\max} \in \{3, 5, 7\}$, $n_{\text{est}} \in \{200, 500\}$, per-target 5-fold CV. Best: C_l uses $\eta = 0.10$, depth 5, 500 trees; C_m uses $\eta = 0.05$, depth 7, 500 trees.

3.4 Multi-Layer Perceptron (DNN)

The `AeroMLP` model (Fig. 2) uses a shared trunk of fully-connected layers with BatchNorm, ReLU, and Dropout, followed by two independent linear heads for C_l and C_m. The shared trunk pushes the model to learn a single geometry–flow representation that works for both outputs at once. BatchNorm keeps activations stable during training, and dropout ($p = 0.1$ in the best config) provides enough regularization without squashing the representations on a dataset this size.

We ran a small architecture search over 20 configurations: five hidden-layer widths ($[128, 128]$, $[256, 128]$, $[256, 256]$, $[256, 128, 64]$, $[512, 256, 128]$), two dropout rates ($p \in \{0.1, 0.3\}$), and two learning rates ($\eta \in \{10^{-3}, 5 \times 10^{-4}\}$), selecting by validation loss. Training uses Adam [8] with MSE loss, a learning-rate scheduler that halves the rate on plateaus, and early stopping after 20 epochs without improvement (max 300 epochs). The best configuration was $[512, 256, 128]$ with $p = 0.1$—wider and shallower beats narrow and deep here, suggesting the dataset needs representational capacity more than regularization.

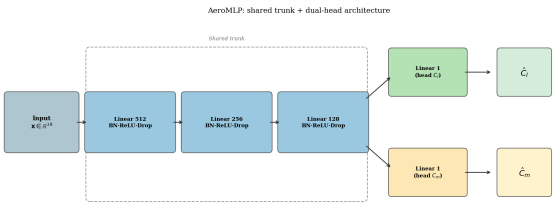

Figure 2: AeroMLP architecture. Shared BN→ReLU→Dropout trunk feeds two independent heads for C_l and C_m, enforcing joint geometry–flow representation learning.

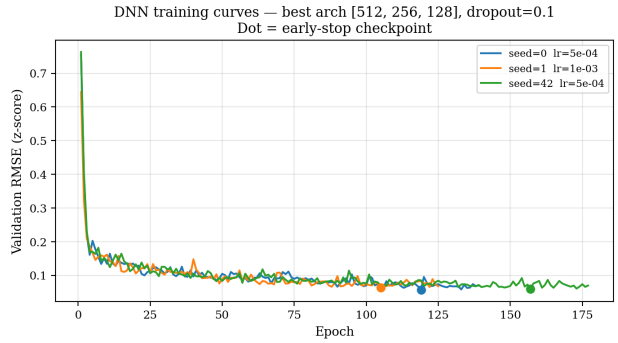

Figure 3: Validation RMSE vs. epoch for three random seeds. Early stopping (patience 20) halts training between 60–120 epochs; the tight seed spread confirms training stability.

To check that results are not seed-sensitive, we train three times with different random initializations (seeds 0, 1, 42). Fig. 3 shows all three converge to similar validation RMSE (0.0571, 0.0631, 0.0594; mean ± std: 0.060 ± 0.003).

4 Results

4.1 Interpolation Performance (7-Airfoil Test Split)

Table 1 shows per-target RMSE and R^2 on the 20% test split. XGBoost achieves the best results ($R^2 > 0.997$); the MLP follows closely; the polynomial baseline performs significantly worse on C_m, where the shock-induced nonlinearity defeats a degree-3 global fit. Despite training on one fewer airfoil, our results match or exceed Elrefaie et al. [1]: they report $R^2 \leq 0.991$ for RF and gradient boosting; we achieve $R^2 = 0.994$ and 0.999 respectively.

Table 1: Interpolation test-split metrics ($N = 238$, 7-airfoil pool)

Model	Target	RMSE	R^2
Polynomial	Cl	0.0311	0.9904
	Cm	0.0269	0.9881
Random Forest	Cl	0.0251	0.9937
	Cm	0.0169	0.9953
XGBoost	Cl	0.0116	0.9987
	Cm	0.0134	0.9971
DNN	Cl	0.0172	0.9970
	Cm	0.0149	0.9964

4.2 Extrapolation to the Held-out RAE2822 Geometry

Table 2 reports metrics on the 172 held-out RAE2822 samples—a supercritical profile entirely absent from training. *All four models yield negative R^2 on both targets.* XGBoost, the best interpolator ($R^2 = 0.9987$), drops to $R^2(C_l) = -0.57$ and $R^2(C_m) = -6.76$; the polynomial baseline reaches $R^2(C_m) = -54.8$. Negative R^2 means predictions are worse than simply predicting the training-set

mean—the surrogate has no useful signal for this geometry.

RAE2822 is a supercritical airfoil with a reflexed camber line that produces a C_m distribution qualitatively different from the NACA profiles in the training set. The 16 surface-coordinate features encode geometry, but not in a form that allows extrapolation to an unseen airfoil family.

Table 2: Held-out RAE2822 metrics ($N = 172$, unseen geometry)

Model	Target	RMSE	R^2
Polynomial	Cl	1.7375	-7.8900
	Cm	1.8501	-54.7682
Random Forest	Cl	1.2363	-3.5008
	Cm	1.2727	-25.3930
XGBoost	Cl	0.7304	-0.5708
	Cm	0.6899	-6.7555
DNN	Cl	1.2491	-3.5948
	Cm	1.2214	-23.3083

4.3 Parity and Error Analysis

Figure 4 shows predicted vs. CFD-true scatter on the interpolation test split. XGBoost and the MLP cluster tightly around the diagonal; the polynomial baseline shows fan-shaped scatter for C_m near $M_\infty = 0.80$; RF falls between the two.

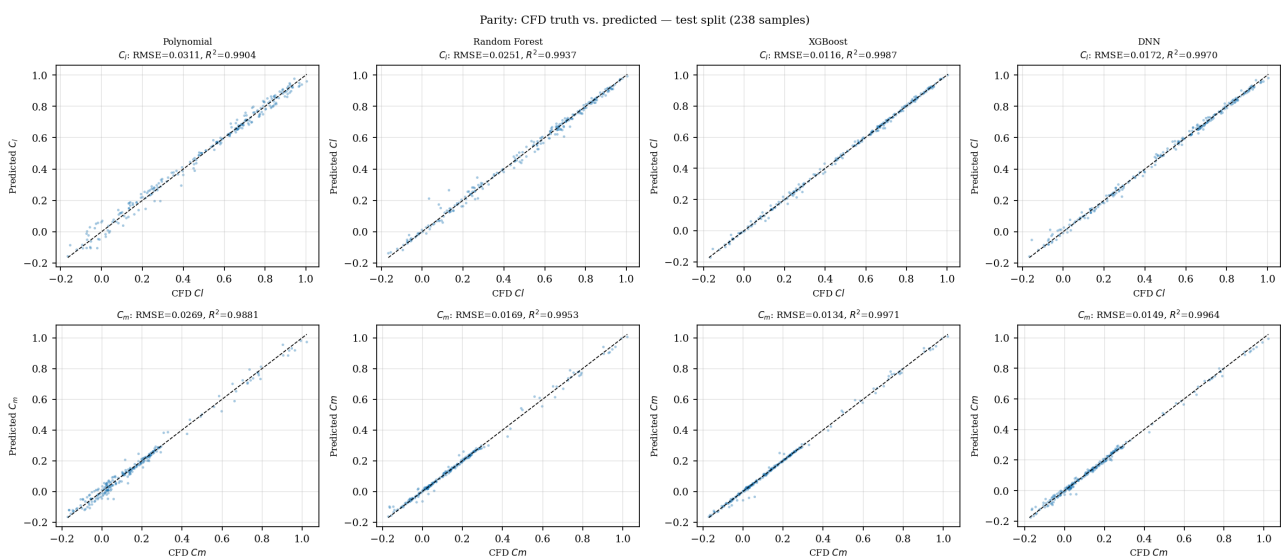

Figure 4: Parity plots for all four surrogates on the test split (238 samples). Rows are C_l (top) and C_m (bottom); columns are models. Dashed line = perfect prediction. RMSE and R^2 annotated per panel.

Figure 5 decomposes RMSE by Mach bin: all models degrade monotonically toward $M_\infty = 0.85$. The polynomial spike at $M = 0.85$ ($\text{RMSE}_{C_l} \approx 0.045$) is $2\times$ its lower-Mach baseline; tree models and the DNN track the same trend at much lower absolute error. Model ranking is *preserved* across all five Mach values, so XGBoost's interpolation lead is not a low-Mach artifact.

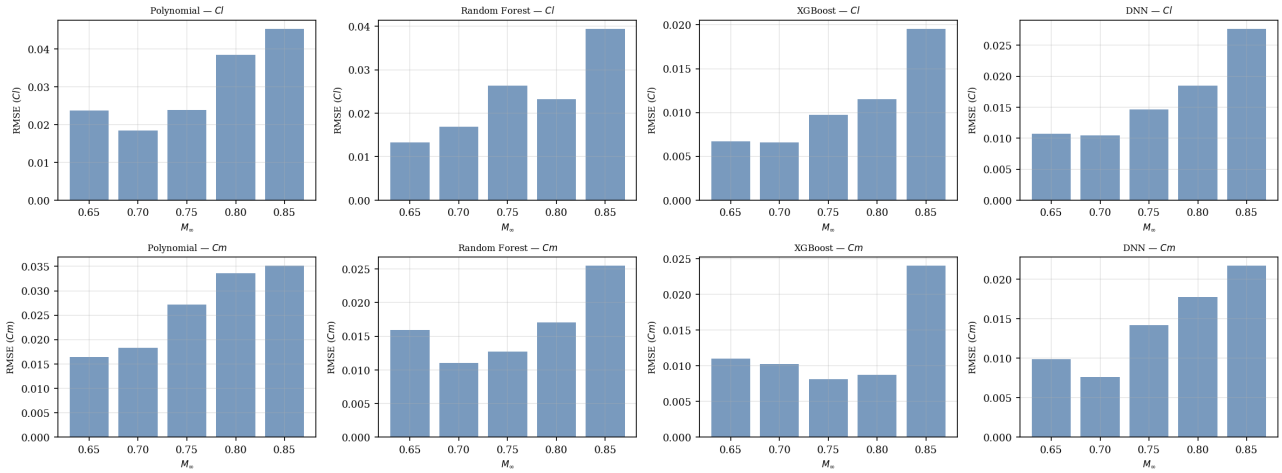

Figure 5: RMSE per Mach bin for each surrogate (test split, 7-airfoil pool). Columns: models; rows: C_l (top), C_m (bottom). Error increases monotonically toward $M_\infty = 0.85$ for all models; the polynomial degradation is the sharpest, and model ranking is consistent across Mach.

4.4 Physical Plausibility and Feature Importance

Figures 6 and 7 overlay predicted $C_m(\alpha)$ at $M_\infty = 0.80$ against CFD ground truth for the held-out RAE2822 profile and show normalized feature importances. The $C_m(\alpha)$ overlay is the hardest physical test: the model must predict the pitching behaviour of an airfoil shape it never saw in training. XGBoost and the MLP traces follow the CFD curve more closely than the polynomial baseline, which falls apart above $\alpha \approx 5°$ where a shock-induced moment shift kicks in that no degree-3 polynomial can follow.

Among all 18 features, α and M_∞ dominate, which is expected—they directly set the shock strength. Of the geometric coordinates, y_{U_1} and y_{U_2} (leading-edge upper-surface curvature) rank highest, consistent with shock formation originating at the leading edge. RF and XGBoost agree on this ranking, which suggests both are responding to the same physical signal rather than dataset-specific noise.

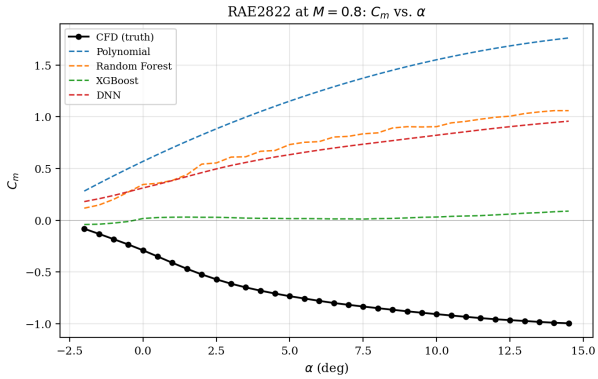

Figure 6: $C_m(\alpha)$ at $M_\infty = 0.80$ for RAE2822. CFD truth (solid black) vs. all four surrogates. XGBoost and MLP track the nonlinear trend; polynomial deviates above $\alpha \approx 5°$.

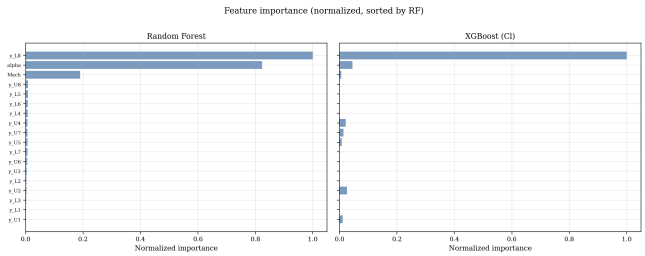

Figure 7: Normalized feature importances from RF (left) and XGBoost (right), sorted by RF rank. α and M_∞ dominate; upper leading-edge coordinates are the most important geometric features.

4.5 Multi-Scenario Model Comparison

Table 3 consolidates interpolation, geometry holdout, Mach extrapolation, and inference latency into a single scorecard. Every model shows the same pattern: near-unity interpolation, catastrophic geometry holdout, moderate Mach extrapolation. This rules out model capacity as the cause—the bottleneck is the training distribution. All four surrogates run a forward pass in $\lesssim 4\,\text{ms}$ vs. $\approx 7\,\text{min}$ for `rhoCentralFoam`, a speedup of 4×10^4–$10^5\times$ regardless of model choice.

Table 3: Per-scenario R^2 and CPU inference latency. Holdout values are negative by design (RAE2822 is out-of-distribution for all models).

5 Extended Analysis

5.1 Leave-One-Airfoil-Out Generalization

To assess how broadly the extrapolation failure generalizes, we run a leave-one-airfoil-out (LOAO) sweep: for each of the eight airfoils, hold it out entirely, retrain the three classical surrogates on the remaining seven with fixed best hyperparameters, and evaluate on the held-out set. Fig. 8 reveals *three* generalization regimes. The 5-digit NACA profiles (NACA23012, NACA24112, NACA25112) cross-generalize well ($R^2 > 0.95$) because the other family members cover the held-out geometry. NACA0012 (only symmetric profile) and NACA4412 (highest 4-digit camber) collapse ($R^2(C_l) = -4.2$ and -6.0); both RAE profiles reach strongly negative R^2. Extrapolation failure is not specific to supercritical profiles—any airfoil whose geometry has no near-neighbor in training will fail.

5.2 Data Efficiency: Learning Curves

Figure 9 plots test-split R^2 vs. training-set size at five fractions (10%–100%), hyperparameters fixed. XGBoost reaches $R^2 > 0.95$ with only 50% of training data (≈ 357 samples), indicating the current dataset is not the primary bottleneck for tree-based models. The polynomial baseline saturates early at a lower ceiling (model capacity, not data). The DNN shows the steepest improvement from 10% to 50%, consistent with neural networks requiring more data to escape underfitting, but converges to near-XGBoost performance at full size.

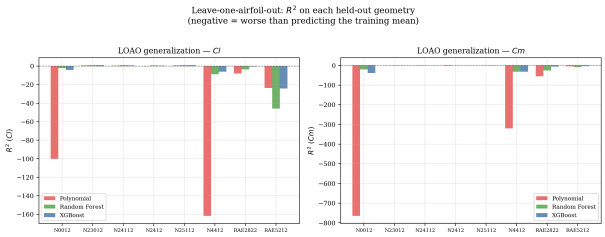

Figure 8: LOAO R^2 per holdout airfoil (three models). Three regimes: 5-digit NACAs cross-generalize well ($R^2 > 0.95$); NACA0012 (symmetric) and NACA4412 (high camber) collapse; both RAE profiles reach strongly negative R^2.

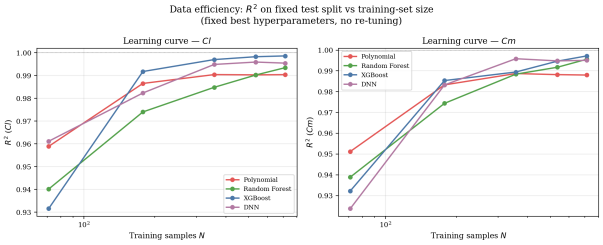

Figure 9: R^2 vs. training-set size (log scale). XGBoost and RF saturate early; the DNN benefits most from additional samples in the 10–50% range.

5.3 Mach-Regime Extrapolation

Training on $M_\infty \in \{0.65, 0.70, 0.75, 0.80\}$ and testing on $M_\infty = 0.85$ (held out entirely) isolates *flight-condition* extrapolation. Unlike the geometry holdout, this does not produce negative R^2: RF and XGBoost achieve $R^2(C_l) \approx 0.90$; the DNN reaches $R^2(C_m) = 0.942$, outperforming XGBoost on that target. Mach extrapolation is a smaller leap—aerodynamic behavior changes smoothly as speed increases, so the step from $M = 0.80$ to $M = 0.85$ is tractable. A new airfoil geometry is a fundamentally different region of input space, which is why that test collapses entirely. The polynomial baseline degrades most ($R^2(C_l) = 0.75$), confirming its inability to handle near-critical shock nonlinearity.

Figure 10 shows the physical basis for the DNN's C_m advantage: tree models saturate at their training-range maximum and flatten out above $\alpha \approx 2°$, while the DNN's continuous representation extrapolates the slope. The polynomial predicts a qualitatively wrong U-shaped curve. Table 4 gives the full per-model breakdown.

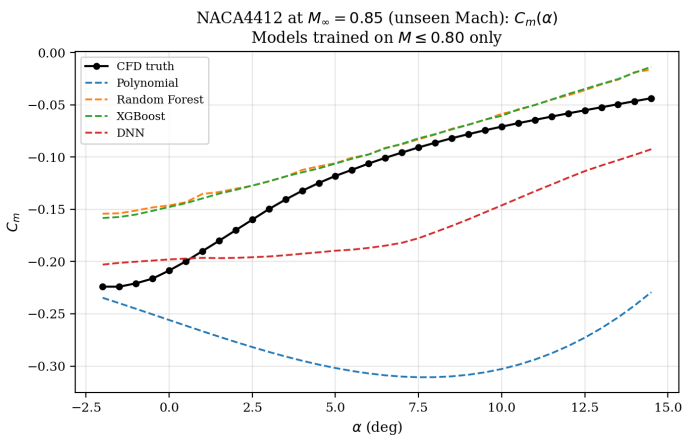

Figure 10: $C_m(\alpha)$ for NACA4412 at $M_\infty = 0.85$ (unseen Mach). Tree models flatten above $\alpha \approx 2°$; the DNN tracks the trend. Polynomial is qualitatively wrong.

Table 4: Mach extrapolation: train $M_\infty \leq 0.80$, test $M_\infty = 0.85$

Model	Target	RMSE	R^2
Polynomial	Cl	0.1689	0.7514
	Cm	0.1476	0.6684
Random Forest	Cl	0.1039	0.9059
	Cm	0.0779	0.9076
XGBoost	Cl	0.1048	0.9044
	Cm	0.0782	0.9070
DNN	Cl	0.1303	0.8522
	Cm	0.0615	0.9424

5.4 Feature Ablation: Full vs. Domain-Selected Features

Comparing the full 18-feature model against the five features from the source paper ($y_{U_1}, y_{U_2}, y_{L_8}, \alpha, M_\infty$), the ensemble and DNN models show essentially no degradation: $\Delta R^2 < 0.001$ for XGBoost and RF. The polynomial baseline is slightly more sensitive ($\Delta R^2 \approx 0.001$–0.003), where the degree-3 basis loses interaction terms between the omitted geometric coordinates. This validates the source paper's feature selection: α, M_∞, and three surface coordinates capture virtually all predictive signal for well-performing models. Table 5 quantifies the per-model ΔR^2.

Table 5: Feature ablation: ΔR^2 (5 features minus 18 features)

Model	$\Delta R^2(C_l)$	$\Delta R^2(C_m)$
Polynomial	-0.0014	-0.0034
Random Forest	-0.0003	-0.0001
XGBoost	-0.0001	-0.0000
DNN	-0.0003	-0.0008

6 Discussion and Conclusion

Model comparison. Table 3 summarises performance across all evaluation scenarios. The polynomial baseline fails where smooth global approximators always fail: its 1,140 degree-3 basis functions cannot represent the shock-induced C_m nonlinearity, and Ridge regularization cannot compensate. XGBoost achieves the best interpolation ($R^2 > 0.999$); fitting separate regressors per target lets C_l use depth 5 and C_m use depth 7—the deeper tree captures the sharper C_m nonlinearity. The DNN provides competitive interpolation and the best Mach-extrapolation C_m ($R^2 = 0.942$ vs. XGBoost's 0.907), and it is the fastest surrogate (0.10 ms) due to vectorised forward evaluation. The Mach-bin residual analysis (Fig. 5) shows that model ranking is consistent across all Mach values, so XGBoost's interpolation lead is not a low-Mach artifact. For interpolation-heavy workloads, XGBoost is the recommended choice; the DNN is preferred when $M > 0.80$ accuracy matters most.

Why the models fail on new geometry. The source paper's $R^2 = 0.996$ is real, but it measures the wrong thing: RAE2822 rows appear in both training and test, so the model only needs to interpolate within geometries it has already seen. Our geometry holdout tests something harder and more practically relevant — can the surrogate predict for a profile it has never encountered? The answer, for every model, is no. RAE2822 is a supercritical airfoil whose C_m curve is qualitatively different from the NACA profiles that dominate training; the 16 surface coordinates encode shape, but the learned mapping is bound to the training distribution and cannot extrapolate across that gap. The LOAO sweep confirms this is not a quirk of RAE2822 specifically: NACA0012 (the only symmetric profile) and NACA4412 (highest camber among 4-digit profiles) collapse just as badly when held out, because they too have geometric properties with no near-neighbors in the remaining training data. The original paper's random-split evaluation never surfaced any of this because RAE2822 rows were always present in training—which means the model never actually had to generalize to a new geometry, it just had to memorize a new operating point on one it already knew.

Limitations and future work. The dataset is inviscid (Euler), 2D, and covers only 8 airfoil families. The core gap exposed here is that a 16-coordinate shape parameterization does not generalize across families, pointing toward geometry-aware encoders (Fourier descriptors, graph networks on the airfoil mesh) as a more transferable representation. Any future benchmark should use geometry-stratified cross-validation rather than random splits. Transfer learning—pre-train on NACA profiles, fine-tune on a handful of new CFD runs—is a practical path to closing the holdout gap without a full retraining sweep.

Acknowledgements

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